

LockDown Lab Walkthrough

TechNova Systems' SOC has detected suspicious outbound traffic from a public-facing IIS server in its cloud platform—activity suggestive of a web-shell drop and covert connections to an unknown host.

As the forensic examiner, you have three critical artefacts in hand: a PCAP capturing the initial traffic, a full memory image of the server, and a malware sample recovered from disk. Reconstruct the intrusion and all of the attacker's activities so TechNova can contain the breach and strengthen its defenses.

PCAP Analysis

Q1 .After flooding the IIS host with rapid-fire probes, the attacker reveals their origin. Which IP address generated this reconnaissance traffic?

Click Statistic tab then dropdown conversation to see the ip generated

Ethernet · 12		IPv4 · 16	IPv6 · 4	TCP
Address A	Address B	Packets	Bytes	
10.0.2.4	10.0.2.15	6,007	4 MB	

This ip 10.0.2.4 creates the most packets with 10.0.2.15

Q2. The attacker is carrying out targeted enumeration against the HTTP service on the IIS host. Based on the HTTP request headers, which tool is being used?

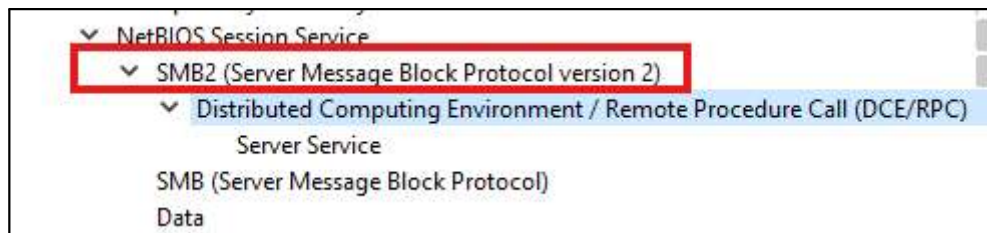
Filter http

2140	94.577381	10.0.2.4	10.0.2.15	HTTP	225	GET	/nmapowercheck1725947084	HTTP/1.1
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Nmap used for enumeration the target against HTTP service

Q3. While reviewing the SMB traffic, you observe two consecutive Tree Connect requests that expose the first shares the intruder probes on the IIS host. Which two full UNC paths are accessed?

Open statistic tab then dropdown click protocol statistics to see lists every single protocol present in the capture



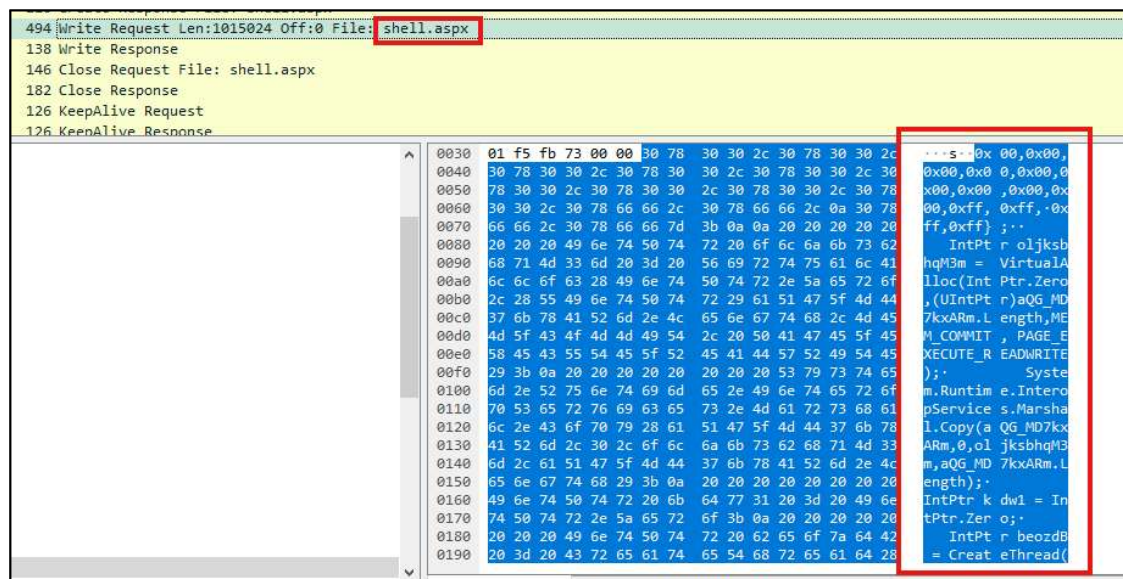
To check the tree connect request in smb filter smb2 which is modern smb2 technology because smb1 WannaCry incident which is insecure.

2671	263.302255	10.0.2.15	10.0.2.4	SMB2	139 Session Setup Response
2672	263.304629	10.0.2.4	10.0.2.15	SMB2	162 Tree Connect Request Tree: \\10.0.2.15\IPC\$
2673	263.305621	10.0.2.15	10.0.2.4	SMB2	138 Tree Connect Response
2674	263.306977	10.0.2.4	10.0.2.15	SMB2	222 Ioctl Request FSCTL_DFS_GET_REFERRALS, File: \\10.0.2.15\Documents
2675	263.307630	10.0.2.15	10.0.2.4	SMB2	130 Ioctl Response, Error: STATUS_FS_DRIVER_REQUIRED
2676	263.309040	10.0.2.4	10.0.2.15	SMB2	126 Tree Disconnect Request
2677	263.309248	10.0.2.15	10.0.2.4	SMB2	126 Tree Disconnect Response
2678	263.312045	10.0.2.4	10.0.2.15	SMB2	172 Tree Connect Request Tree: \\10.0.2.15\Documents
2679	263.316785	10.0.2.15	10.0.2.4	SMB2	138 Tree Connect Response

\\10.0.2.15\IPC\$

\\10.0.2.15\Documents

Q4. Inside the share, the attacker plants a web-accessible payload that will grant remote code execution. What is the filename of the malicious file they uploaded?



Why this file upload is suspicious because of the blob which show some obfuscated script

Memory Dump Analysis

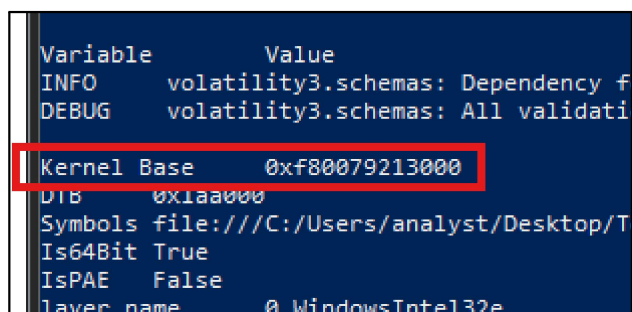
Q6. Your memory snapshot captures the system's kernel in situ, providing vital context for the breach. What is the kernel base address in the dump?

```
function vol { py "C:\Users\analyst\Desktop\Tools\volatility3-develop\vol.py" @args }
```

//to make each command write only vol, configure this setting in \$PROFILE

```
vol -f "C:\Users\analyst\Desktop\memdump.mem" windows.info
```

windows.info = Show OS & kernel details of the memory sample being analyzed.



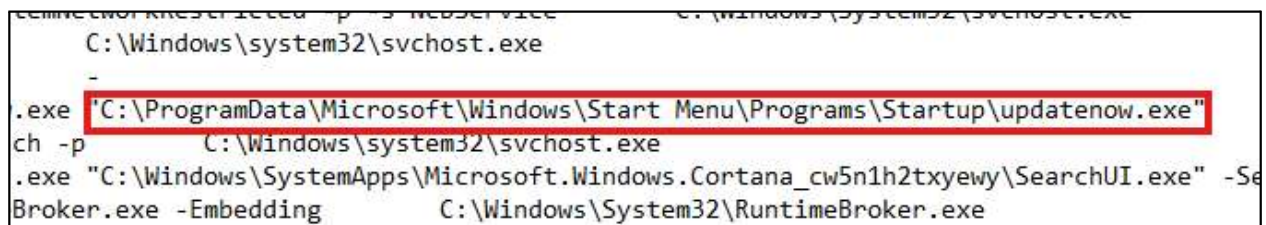
```
Variable      Value
INFO          volatility3.schemas: Dependency f
DEBUG         volatility3.schemas: All validati
Kernel Base    0xf80079213000
DIB           0x1a0000
Symbols file:///C:/Users/analyst/Desktop/T
Is64Bit       True
IsPAE         False
Layer name     0 WindowsIntel32e
```

0xf80079213000

Q7. A trusted service launches an unfamiliar executable residing outside the usual IIS stack, signalling a persistence implant. What is the final full on-disk path of that executable?

```
vol -f "C:\Users\analyst\Desktop\memdump.mem" windows.pstree |
  Out-File "C:\Users\analyst\Desktop\pstree.txt"
```

Check .pstree to quickly check malicious process



```
cmd.exe C:\Windows\system32\svchost.exe
-
.exe "C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe"
ch -p C:\Windows\system32\svchost.exe
.exe "C:\Windows\SystemApps\Microsoft.Windows.Cortana_cw5n1h2txyewy\SearchUI.exe" -Se
Broker.exe -Embedding C:\Windows\System32\RuntimeBroker.exe
```

C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe

Why suspicious because the .exe file is stored in system-wide startup folder, files execute when the user login.

Also it imitates a legitimate updater name.

Q8.. The reverse shell's outbound traffic is handled by a built-in Windows process that also spawns the implanted executable. What is the name of this process, and what PID does it run under?

```
**** 900      4332      updatenow.exe  0x
```

The parent process id is 4332 of the .exe file which we will trace it running volatility netscan to see outbound traffic.

vol -f "C:\Users\analyst\Desktop\memdump.mem" windows.netscan

```
0xce0656fbd050 UDPv6  :: 0 * 0 776 svchost.exe 2024-09-10 05:30:01.000000 UTC
0xce0657209270 TCPv4  10.0.2.15 49688 10.0.2.4 4443 CLOSED 4332 w3wp.exe 2024-09-10 05:49:51.000000 UTC
0xce0657271140 UDPv4  0.0.0.0 0 * 0 2440 svchost.exe 2024-09-10 05:30:17.000000 UTC
```

W3wp.exe, 4332

Malware Sample Analysis

Q9. Static inspection reveals the binary has been packed to hinder analysis. Which packer was used to obfuscate it?

```
PS C:\Users\analyst\Desktop > file .\updatenow.exe
.\updatenow.exe: PE32 executable (GUI) Intel 80386, for MS Windows, UPX compressed
```

The packer is UPX

Q10. Threat-intel analysis shows the malware beaconing to its command-and-control host. Which fully qualified domain name (FQDN) does it contact?

```
PS C:\Users\analyst\Desktop > upx -d .\updatenow.exe
Ultimate Packer for Executables
Copyright (C) 1996 - 2025
UPX 5.0.1 Markus Oberhumer, Laszlo Molnar & John Reiser May 6th 2025

File size      Ratio      Format      Name
-----
1084928 <- 602112 55.50% win32/pe updatenow.exe
```

Unpack the file with upx tools

```
PS C:\Users\analyst\Desktop > file .\updatenow.exe
.\updatenow.exe: PE32 executable (GUI) Intel 80386, for MS Windows
```

This file is a window executable file


```

PS C:\Users\analyst\Desktop > Get-FileHash .\updatenow.exe -Algorithm SHA256

Algorithm      Hash
-----
SHA256         91962E35BA555E81C829A6784A73915D392361FC5E9F98807CEBDC238DBCE6D3
Path
-----
C:\Users\analyst\Desktop\updatenow.exe

FLARE-VM 07/16/2026 18:13:39
PS C:\Users\analyst\Desktop >

```

Get the file hash

When it comes to threat intelligence, virus total is one of the platform

91962e35ba555e81c829a6784a73915d392361fc5e9f98807cefdc23bdbce6d3

wypakowany.exe

peexe detect-debug-environment checks-bios calls-wmi checks-network-adapters

53 / 72 Community Score

53/72 security vendors flagged this file as malicious

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 1

[Join our Community](#) and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

Contacted Domains (1)

Domain	Detections	Created	Registrar
cp8nl.hyperhost.ua	2 / 91	-	ua.ukrnames

Go to the relations tab and it shows the domain.

Q11. Open-source intel associates that hash with a well-known commodity RAT. To which malware family does the sample belong?

The screenshot shows the VirusShare community page for a specific file hash. At the top, a red circle indicates a community score of 53/72. Below this, a red banner states "53/72 security vendors flagged this file as malicious". The file name is "wypakowany.exe". A list of tags includes "peexe", "detect-debug-environment", "checks-bios", "calls-wmi", and "checks-network-adapters". The "COMMUNITY" tab is selected, showing a comment from "ANY.RUN" dated 8 months ago. The comment contains a "Sandbox Analysis" with a "Verdict: Malicious activity" and "Tags: stealer, ultravnc, rmm-tool, agenttesla". A red box highlights the "HTML Report" link: <https://any.run/report/91962e35ba555e81c829a6784a73915d392361fc5e9f98807cebd23bdbc6d3/7c24cbf2-29ff-41b3-a114-02558396ac56>. A "Watch full analysis" link is also present.

Go to community tab and open the any run report

The screenshot shows the "General Info" section of the ANY.RUN report. It lists the "File name" as "updatenow.exe" and the "Full analysis" URL as <https://app.any.run/tasks/7c24cbf2-29ff-41b3-a114-02558396ac56>. The "Verdict" is "Malicious activity". The "Threats" section lists "Agent Tesla", "Agent Tesla", and "Stealer". A red box highlights the "Agent Tesla" threats. Below this, a description states: "Agent Tesla is spyware that collects information about the actions of its victims by recording keys dedicated website where this malware is sold."

This is the answer

The screenshot shows the question "Q11" with a green checkmark, a weight of 2, and 1971 solves. The question text is "Open-source intel associates the sample belong?". Below the question, the answer "Agent Tesla" is shown in a text input field.

Agent Tesla is commonly described as a commodity RAT and information stealer. It can collect credentials, keystrokes, clipboard data, browser information, and other victim data.